



A polyomino is called *rectifiable* if there is some number of copies of it that will tile a rectangle.

We say that a polyomino has *rectifiability defect* n if we can tile a rectangle missing n holes. (If the polyomino is rectifiable, this metric is 0.)

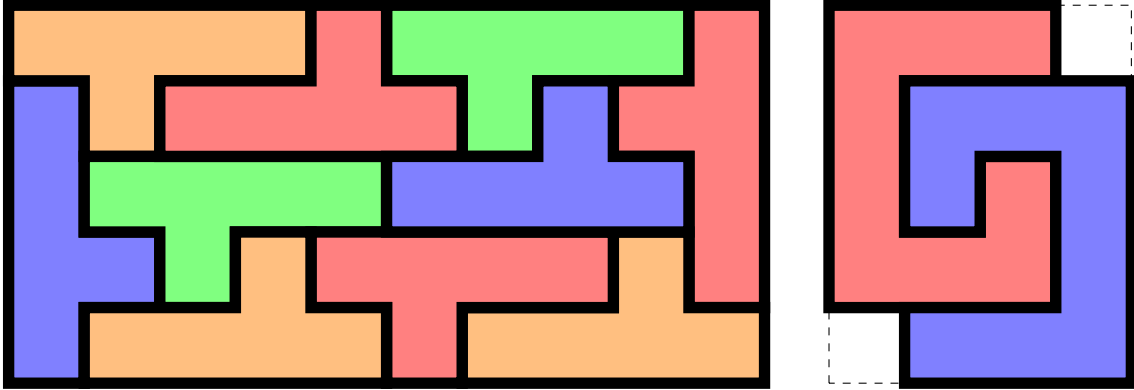


Figure 1: An example showing a rectifiable 5-omino, and a non-rectifiable polyomino that has rectifiability-defect of 2.

Question. Let $\text{grd}(n)$ be the greatest rectifiability-defect over all n -ominoes. What can you say about $\text{grd}(n)$?

Related.

1. Related question.
2. If instead, we can have the non-rectifiable polyominoes overlap, what is the least amount of overlap required to tile a rectangle?
3. What if instead of tiling rectangles, we want to tile squares? Or tile scaled up versions of the polyomino?

References.

Problem 138.

Peter Kagey, Nerd sniped by Richard Stanley

Richard Stanley, What are the most attractive Turing undecidable problems in mathematics?